Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Repository Link / URL	https://github.com/xxxxxx/credit-card-approval-prediction
Programming Language(s)	Python, JavaScript, SQL
Framework(s) Used	FastAPI (Backend), React (Frontend), Tailwind CSS
Key Features Implemented	End-to-end ML classification pipeline using XGBoost. SHAP integration for Explainable AI (XAI) reports. REST API for real-time model predictions. Interactive Agent Dashboard for data entry and result visualization. Automated data preprocessing (handling missing values, scaling).
Pending / Incomplete Features	Automated retraining pipeline triggered by data drift. Advanced cloud-based logging integration.

Setup / Run Instructions	**1. Backend:** Navigate to `/backend`, run `pip install -r requirements.txt`, then start the server using `uvicorn main:app --reload`. **2. Frontend:** Navigate to `/frontend`, run `npm install`, then start the client using `npm start`. **3. Database:** Ensure PostgreSQL is running locally on port 5432 and run database migrations.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The machine learning model was successfully integrated into the backend REST API. The solution thoroughly addresses the core problem of "black-box" credit rejections by actively passing the calculated SHAP explainability values directly to the frontend for bank agents to view, allowing for complete transparency in the approval process.